

ADRENAL GLAND DISEASES — Clinical Revision Table

Bold = high-yield. SYMPTOMS column = standard clinical knowledge added (not in lecture). Glands: cortex → cortisol/aldosterone/androgens; medulla → catecholamines.

CORTISOL EXCESS — CUSHING'S SYNDROME

Disease	Cause / pathophysiology	SYMPTOMS & SIGNS (added)	Investigations	Management
Cushing's syndrome (general)	Cortisol excess. Commonest = iatrogenic steroids (even inhaled/topical). Endogenous: pituitary (disease), adrenal tumour, ectopic ACTH. F:M 4:1 (ectopic = M)	Moon face, plethora, central obesity, buffalo hump, supraclavicular fat. Purple striae, easy bruising, thin skin. Proximal myopathy (can't rise from chair). Hirsutism, acne. HTN, weight gain. Mood change/psychosis, depression. Oligo/amenorrhoea, ↓libido. Osteoporosis, glucose intolerance/DM, poor healing, infections	2-step. Confirm (2 of 3): fail LDDST (0.5mg QDS 48h), ↑late-night cortisol (lost circadian), ↑24h UFC. Then ACTH: undetectable=adrenal ; high=pituitary/ectopic. HDDST suppresses pituitary not ectopic; MRI; BIPSS; CT adrenal	Untreated severe = 50% 5-yr mortality . Surgery mainstay. Medical: metyrapone, ketoconazole, osilodrostat (titrate to cortisol)
Cushing's DISEASE (pituitary)	Pituitary ACTH-secreting microadenoma (<10mm). Retains some feedback	As above + possible secondary hypogonadism. Macroadenoma mass effects rare	ACTH high; suppresses on HDDST; +CRH stim; MRI ~60% sensitive; BIPSS confirms	Trans-sphenoidal surgery (70% remission, 20% recur → lifelong follow-up). Bilateral adrenalectomy → Nelson's . Pasireotide
Ectopic ACTH	Non-pituitary ACTH (lung small-cell, bronchial carcinoma). No feedback → ↑↑ACTH+cortisol	Marked pigmentation, hypokalaemic alkalosis, severe myopathy, hyperglycaemia, oedema. Classic Cushing features LESS common; rapid + cachexia if malignant	ACTH high, NOT suppressed by HDDST; CT chest/abdomen to find tumour	Remove tumour (carcinoid). Incurable → medical or bilateral adrenalectomy
Adrenal tumour	Cortisol-secreting adenoma or carcinoma (>5cm, malignant features)	Cushingoid features; adenoma slower, carcinoma rapid ± abdominal mass/pain	ACTH undetectable ; CT/MRI adrenal	Laparoscopic adrenalectomy (adenoma). Carcinoma: surgery+radiotherapy+ mitotane (poor prognosis)

CORTISOL/ALDOSTERONE DEFICIENCY — ADRENAL INSUFFICIENCY

Disease	Cause / pathophysiology	SYMPTOMS & SIGNS (added)	Investigations	Management
Adrenal insufficiency (general)	↓cortisol ± aldosterone. Commonest = ACTH deficiency (secondary) from steroid withdrawal / pituitary tumour. Addison's + CAH rarer	Fatigue, weakness, weight loss, anorexia, N&V, abdominal pain, dizziness/postural hypotension, salt craving, myalgia. Often misdiagnosed as depression/CFS	Suspect if unexplained fatigue, hyponatraemia, hypotension	Lifelong replacement after stabilization
Addison's (primary)	Autoimmune (21-hydroxylase Ab) commonest; also TB, mets, HIV. Eventually fails GC+MC	All above + PIGMENTATION (palmar creases, buccal, scars) from ↑ACTH/MSH. Vitiligo 10-20%. Weight loss uniform. May have other autoimmune disease (APS I/II)	Short Synacthen test (cortisol fails to rise). ACTH HIGH. 21-OH Ab; supine renin↑/aldo low; CT if Ab-ve	Hydrocortisone 15-20mg/d + fludrocortisone 0.05-0.15mg/d (MC, primary only). ± DHEAS women
Acute adrenal crisis	Precipitated by illness, surgery, infection in known/unknown insufficiency	Circulatory SHOCK, severe hypotension, hypoNa, hyperK, hypoglycaemia ± hyperCa. Cramps, N&V, diarrhoea, fever, confusion	TREAT BEFORE LABS. Store cortisol/ACTH	IV hydrocortisone + IV saline/glucose immediately. No delay
Secondary (ACTH deficiency)	Steroid withdrawal (commonest), pituitary disease. HPA recovery ≥1 yr	Same as primary BUT NO pigmentation (low ACTH) & NO hyperkalaemia (aldosterone intact)	Short Synacthen fails; ACTH LOW	Hydrocortisone only — fludrocortisone NOT needed. Alternate-day steroids ↓suppression

ALDOSTERONE EXCESS — PRIMARY HYPERALDOSTERONISM

Disease	Cause / pathophysiology	SYMPTOMS & SIGNS (added)	Investigations	Management
Primary hyperaldosteronism	Up to 10% of HTN. Most = bilateral adrenal hyperplasia ; minority = APA (Conn's) . Rare: GC-suppressible (AD)	Often asymptomatic. HTN (resistant). Hypokalaemia → muscle weakness/paralysis (esp SE Asian), polyuria/nocturia, cramps, tetany. Oedema (Na retention). Accelerated HTN rare	Test if hypokalaemia/resistant HTN/young/FH. ↑ARR (aldosterone:renin) . Bloods: hypokalaemic alkalosis, Na high-normal. Confirm: low renin+high aldo → saline suppression. CT/MRI; adrenal vein sampling	MR antagonist: spironolactone (gynaecomastia 20%) or eplerenone; amiloride. APA → laparoscopic adrenalectomy (HTN persists ~70%)

CATECHOLAMINE EXCESS — PHAEOCHROMOCYTOMA / PARAGANGLIOMA

Disease	Cause / pathophysiology	SYMPTOMS & SIGNS (added)	Investigations	Management
Phaeo / paraganglioma	Catecholamine-secreting NET. 80% adrenal medulla, 20% extra-adrenal. 15% malignant. 40% inherited (NF, VHL, MEN 2a/2b, SDH genes)	Classic triad: episodic HEADACHE + SWEATING + PALPITATIONS. Paroxysmal/sustained HTN, pallor, anxiety/panic, tremor, chest/abd pain, weight loss. Postural hypotension between attacks	Plasma/urine metanephrines + normetanephrines (false+ in stress/TCA → repeat). Chromogranin A. Genetic test if <50/FH. CT/MRI; MIBG scintigraphy; DOTATATE-PET	Pre-op ≥6 wks: α-blocker (phenoxybenzamine) FIRST , then β-blocker if tachycardia. NEVER β before α (paradoxical ↑BP). Surgery; intra-op nitroprusside/phentolamine

ENZYME DEFECT — CONGENITAL ADRENAL HYPERPLASIA (CAH)

Disease	Cause / pathophysiology	SYMPTOMS & SIGNS (added)	Investigations	Management
CAH (21-OH deficiency)	Autosomal recessive. Cortisol enzyme block → ↓feedback → ↑ACTH → hyperplasia + androgen excess. 21-hydroxylase = commonest → ↑17-OH-progesterone	Severe (1/3, infancy): ambiguous genitalia (girls), salt-wasting crisis (vomiting, dehydration, shock). Milder (2/3): precocious pseudo-puberty (boys), virilization. Late-onset: adult female hirsutism + amenorrhoea	↑17-OH-progesterone. ↑androgens, low gonadotrophins (distinguishes pseudo from true precocious). Renin (salt-wasting). 17/11β-OH deficiency → HTN (11-deoxycorticosterone)	Replace corticosteroids + suppress ACTH-driven androgens (balance vs Cushingoid). Children: monitor growth velocity. Late-onset: anti-androgen if hirsutism main issue